

Empty Vessels Make Waves!

A Bangalore student's improvised Wi-Fi network is inspiration for the do-it-yourself-er in all of us!

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There are a few people who won't let a technology or resource limitation cow them down. They try and tweak with vengeful perseverance until they find a way out. Meet Ravi Sharma, a final year B.Com. student in Bangalore who has set up a Wi-Fi network using low-cost TV cables and a simple yet brilliant directional antenna made from trashed utensils!

Making one's own antenna may not be unusual for Ham radio veterans, but a lousy cable and used utensils for a Super G Wi-Fi computer network? Yes, that's what this soft-spoken 20-year-old has done!

The Challenge

Like they say, necessity is the mother of all inventions: in Ravi's case, he wanted to network the computers in his father's drug distribution godown and the administration office, which are about a hundred metres apart.

A conventional wired LAN was out of the question, since the area is thickly populated—any kid with a blade or a pair of scissors in his itchy hands could bring the network down. The next thing to consider was to go the wireless way.

Cost immediately became a concern because of the distance between the buildings, and also because of the distance between the godown computer and the terrace from where line-of-sight could be achieved. Ravi could not afford commercial setups that cost upwards of Rs 50,000. He therefore started trying to figure how to build his own setup.

The Solution

To keep costs low, Ravi used an ordinary RJ58 TV cable, which costs about Rs 10 per metre—the typical low-loss cables cost about Rs 400. This in itself is some serious cost cutting! With such a cable, he was able to get just about enough signal strength to take data to an access point located 50 metres away on the terrace. From here, the access point had to talk to the WLAN card on the other machine 100 metres away. Ravi could not get this working immediately with the omni-directional antenna that came with the WLAN card. Rather than invest in a commercial high-gain directional antenna, he thought of making one himself.

He figured that all he needed was an old aluminium utensil. He placed the antenna in front of the inner curve of the utensil. Thus was formed a brilliant yet simple zero-cost directional antenna with an estimated 20 dB gain! Ravi similarly placed another scrapped aluminium vessel behind the 5 dB antenna of the access point. The range was increased by at least 50 per cent with the use of the vessels, which act as reflectors. He has also weather-proofed them by applying a coat of an anti-rust paste.

Getting his setup working was difficult—first, he could not find a connector to connect the RJ58 cable to the WLAN card's antenna port. At a point where many would have given up and opted for more expensive cables, Ravi again made one himself! Mind you, that's no easy task! With the cable connected at one end, it was only half the problem solved. To attain line-of-sight, he had to place the antenna high on his terrace, and doing this needed some serious wall climbing skills. In his own words, "The biggest challenge was to climb

